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How CEOs use Twitter: A comparative analysis of Global and Latin American companies

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ABSTRACT

Social networks are becoming a key communication tool for organizations, but also for top managers like CEOs. Among the different available platforms, Twitter is one of the greatest and it is considered one of the most suitable to share information and engage in dialogue with stakeholders. In this way, this paper analyzes the presence of CEOs on the most active social network sites, and assess the activity and interaction of these top managers on Twitter. CEOs from Global and Latin American companies were selected, to compare their performance. The results of the study show that the presence of CEOs in social networks is very low, and the majority of those that are present on them are not adequately using their Twitter accounts. Although the general presence and performance on are low, LatAm CEOs have a better presence on social networks and they are more active on Twitter, but Global CEOs have better interaction results on their accounts. So, this area of strategic communication should be improved by communication practitioners, since the CEO communication is nowadays a key communication issue for any organization.

1. Introduction

During the last decade, the social network sites (SNS) have become an important field of the communication management, since companies or institutions recognize that these tools are key tools for reaching their communication objectives (Evans, Twomey, & Talan, 2011; Men & Tsai, 2016).

The SNS has also become a good channel to develop and improve the leadership of top management, especially for CEOs. Twitter is one of the most used SNS because it is considered a good dialogical tool for getting in touch in real time with customers, employees, media and the general public (Weber Shandwick, 2012). It helps to improve the CEOs reputation and benefits the company by showing their human side (Brandfog, 2016; Domo & CEO.Com, 2015; Weber Shandwick, 2017). Although SNS like Twitter gives significant benefits, current research highlight that the number of CEOs that have an account in any social media is low and some leaders keep on despising its importance (Porter, Anderson, & Nhotsavang, 2015).

Then, the general objective of this research is to analyze if CEOs of Companies have a presence in SNS, and how they are using Twitter as a communication tool. In addition, a comparison will be made between CEOs of Latin American (LatAm) and Global companies, in order to know if there are similarities or differences between them.

2. Literature review

2.1. The CEO's communication: an organizational communication issue

In last years, the social media has acquired an important role in the communication strategies of many institutions and companies (Tench, Moreno, Navarro, & Zerfass, 2015; Tench, Verčič, Zerfass, Moreno, & Verhoeven, 2017). Social media has created a demand for real-time, two-way communication programs that develop relationships and engage publics based on the conversation (Lewis & Nichols, 2010; Scott, 2010). Communication practitioners have also embraced social media as an important tool to enhance issues management, environmental monitoring and to identify the "faith-holders" and "hate-holders" that a company has in the social media arena (Johansen, Johansen, & Weckesser, 2016; Lewis & Nichols, 2010; Wright & Hinson, 2009, 2010), since these stakeholders are using social media to share, create, modify and discuss content, as well as to buy products or services (DiStaso & McCorkindale, 2013; García Galera, Fernández Muñoz, & Porto Pedrosa, 2017; Kietzmann, Hermkens, McCarthy, & Silvestre, 2011; Zamora Medina, Sánchez Cobarro, & Martínez Martínez, 2017). Its weight has increased in such way that some authors go so far as to affirm that the social networks are changing the way of communicating in organizations (Losada-Díaz & Capriotti, 2015; Van den Berg &

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Verhoeven, 2017).

Companies from the Fortune 500 ranking are using Facebook or Twitter to connect, disseminate information and strive to maintain a dialog with stakeholders and generate engagement (Rybalko & Seltzer, 2010). According to DiStaso and McCorkindale (2013), the 74% of the 250 Fortune's Most Admired U.S. Companies have a YouTube account. The same study also shows that most companies adopted social media and have at least an account on Facebook, Twitter or YouTube (only the 9% of companies do not have an account on these SNS). These tendencies of using social media can also be identified in other institutions like museums (Caerols-Mateo, Viñarás-Abad, & Gonzálvez-Valles, 2017; Capriotti, Carretón, & Castillo, 2015), hospitals (Rando Cueto, Paniagua Rojano, & de las Heras Pedrosa, 2016), mass media (Rivera Rogel, Carrión Salinas, & Córdova Camacho, 2016) and NGO organizations like the American Red Cross (Briones, Kuch, Liu, & Jin, 2011) or even in city branding (Huertas, Setó-Pàmies, & Míguez-González, 2015).

Like any other corporate communication issue, the CEO communication activities must be monitored and managed as a part of overarching communication strategies that embrace internal and external communication, both symbolically and behaviorally (Christensen and Cornelissen, 2011; Grunig, 1993; R. Edelman, 2017; Zerfass, Verčič, & Wiesenber, 2016). Zerfass et al. (2016) affirm that CEO positioning and communication will be more successful and sustainable if it is based on a sound management process.

Considering the CEO as the public "face" of the company, Men (2012) asserted that the CEO strongly contributes to defining the organization's reputation to internal and external stakeholders. The CEO's opinion, even though it is not about companies' core business, can call the attention of the public opinion and it could affect the company in a positive or negative way (Chatterji & Toffel, 2017; George, 2017; R. Edelman, 2017). Facing external publics, CEOs are corporate spokespersons who are actively visible and favorably shaping the corporate reputation (Men, 2012). Internally, CEOs influence employee perception, attitudes, and performance through leadership and power (Men, 2012; Park & Berger, 2004).

However, the credibility of leadership is in crisis: only the 37% of CEOs are considered very/extremely credible (Edelman, 2017). Of the 28 countries surveyed, in 23 countries CEOs are not credible, and in all of these countries, their credibility has declined (Edelman, 2017). These figures are not encouraging but reveal something relevant: the CEO's communication should continue being worked by the communications departments more than ever. It is necessary to have a clear decision-making protocol that can be implemented strategically (R. Edelman, 2017). In this way, the research findings of Zerfass et al. (2016) reflect that there is a large gap between striving for CEO positioning and the management practice of CEO positioning and communication. For this reason, these authors point out that there is a lack of professionalism and a weak grade of institutionalization in this field of communication.

2.2. The CEOs and the social media

One decade ago, Terilli and Arnorsdottir (2008) analyzed the CEO's blogs, and they found that the words of a CEO can affect markets, share prices, communities, employees, and consumers. Talking about the role of CEOs on social media, some recent research (Edelman, 2014; Holtz, 2014; Men & Tsai, 2016) confirm this idea, and point out that CEOs are stepping up from the conventional role of corporate spokesperson, and they are now actively involved in organizational communication strategies. Several studies show that the presence of a CEO in social networks benefits the organization and the own CEO's reputation, principally because:

- It shows the personal, human side of the company and the CEO (Domo & CEO.Com, 2015; Weber Shandwick, 2012), and publics have the possibility to get more direct and open interactions with corporate leaders on social media (Men & Tsai, 2016).
- Social media engagement makes CEOs more effective leaders. Helps to build good relationships with the news media and makes them more effective in managing crises (Brandfog, 2016; Weber Shandwick, 2012).
- Makes a brand more honest and trustworthy: the social CEOs have a positive impact on the perception that stakeholders have about the brand, tending to enhance its credibility in the market and be considered more transparent (Brandfog, 2016; Weber Shandwick, 2012).
- Helps to associate the company with some features like innovation, competition and get positive impacts in business results (Brandfog, 2016; Domo & CEO.Com, 2015; Weber Shandwick, 2012).
- Helps to connect with current and potential customers on a more personal level and gives the possibility of understanding customers better through more direct engagement and social listening. In this way, gather significant customer loyalty (Brandfog, 2016; Domo & CEO.Com, 2015; Weber Shandwick, 2012; Weber Shandwick, 2017).
- It is a good way for CEOs to communicate directly with employees and helps them understand and stay in touch with what is happening inside the company (Brandfog, 2016; Weber Shandwick, 2012; Weber Shandwick, 2017).
- A company with a human face of authority and responsibility (i.e., CEO) contributes to crisis damage control and reviving of the company after a crisis (Turk, Jin, Stewart, Kim, & Hipple, 2012).
- Attract and recruit top talent to work for the company and make it a more attractive place to work (Domo & CEO.Com, 2015; Weber Shandwick, 2012).
- It is a good way of sharing news and information about the company (Weber Shandwick, 2012).

2.3. Twitter as a valuable tool for CEO communication and leadership

Twitter has become a valuable communication tool for brands and organizations (Deltell, Osteso, & Claes, 2013; Lim & Lee-Won, 2017; Van den Berg & Verhoeven, 2017). Some studies show that Twitter as the very dialogical social network and for that reason is chosen by companies and customers to get in contact (Evans et al., 2011; Rybalko & Seltzer, 2010). It allows provide quick updates, opinions or information and facilitates a real-time communication between members (Aladwani, 2015; Evans et al., 2011). In this sense, they believe that Twitter offers a form of communication not offered by other social media applications, and they believe microblogging will continue to be an essential part of an integrated communications campaign (Evans et al., 2011).

Evans et al. (2011) findings suggest that communication practitioners consider microblogging to be a valuable asset to a campaign's social media strategy. The 78% of the companies on the 2015 Fortune 500 list maintain an active Twitter account and more than half of the fastest-growing private U.S. companies listed on the 2013 Inc. 500 pointed to Twitter as having the greatest potential to drive sales growth in comparison with other social media platforms (Barnes & Lescault, 2014; Barnes, Lescault, & Holmes, 2015; Lim & Lee-Won, 2017). Twitter adoption is strong, not only among for-profit organizations but also among non-profit organizations, with 86% of the top 400 charities and nonprofits on the 2013 Philanthropy 400 list using Twitter for public relations (Barnes, 2014; Lim & Lee-Won, 2017).

The research made by Evans et al. (2011) about Twitter in the communication and public relations field shows that industry leaders view Twitter as a valuable asset to their daily practice. The nature of Twitter communications provides practitioners with the ability to create a conversation and a relationship with their audience in real time. Moreover, allows them to micro-target communications to specific communities and opens a one-to-one line of communication with users of the media (Evans et al., 2011). Rybalko and Seltzer (2010) assert that SNS such as Twitter would seem to be capable of providing

an organization with a wide range of opportunities to engage their publics in dialogic communication.

Although SNS like Twitter gives significant benefits, current studies highlight that the number of CEOs that have an account in any social media is low and some leaders keep on despising its importance (Porter et al., 2015). Being on social networks could help to improve the CEOs reputation and benefits the company by showing their human side (Brandfog, 2016; Domo & CEO.Com, 2015; Weber Shandwick, 2017).

Different leaders around the world choose Twitter to make important announcements, support social movements or other leaders, connect with their followers, or simply to show some aspects of their personal life. It is possible to find politicians, religious leaders, and CEOs using Twitter accounts. Some studies analyze the use of Twitter by different leaders, like politicians (Aharony, 2012; Frame & Brachotte, 2015; López de Ayala-López, Catalina-García, & Fernández-Fernández, 2016). There are rankings that mention who The Top CEOs on Social Media (Hootsuite, 2016) were, and Church Leaders has a ranking with the Top Christian Leaders to Follow on Twitter (Church Leaders, 2013).

Although the number of CEOs in Twitter is significantly growing, the percentages of presence remain relatively low: only 10% of CEOs from Fortune 500 companies had a Twitter account in 2015 (Domo & CEO.Com, 2015). The number of CEOs that have an account in any social media is low and some leaders keep on despising its importance (Porter et al., 2015). Some recent research points out that in the case of CEOs using Twitter, they have a low credibility and value of social media and tend to engage in one-sided conversations in a two-way medium (Porter et al., 2015). Some authors (Men & Tsai, 2016; Porter et al., 2015) argue that this aspect of the CEOs communication should continue being analyzed, bearing in mind that it is a tendency in growth.

3. Methodology

The general objective of this research is to analyze if CEOs of Companies have a presence in SNS, and how they are using Twitter as a communication tool. So, the research object of this study is the Twitter accounts of the CEOs of companies, and Global and Latin American (LatAm) businesses companies were selected, with the intention to compare both of them.

In order to select which CEOs could be analyzed, it was considered two of the most representative rankings: Fortune Global 500 (for global companies) and América Economía 500 (for Latin American companies). In both cases, companies are ranked by total revenues for their respective fiscal years ended and are considered the 500 largest businesses. Rankings of 2015 were selected (published in middle 2016).

After CEOs of these companies were identified, it was necessary to take out some companies from the rankings, because they were in both rankings at the same time. For those who were from Latin America and were ranked also in Fortune, they were taken out from Fortunes' ranking. On the other hand, when some global companies were ranked in América Economía, first was analyzed if these companies have a regional or country CEO or if it was the same global CEO. When there was a regional or country CEO, this person was taken into account and the company was left in both rankings. If the CEO was the same as the global company, it was taken out of the América Economía' list.

In the Global CEO list, the number of CEOs is bigger than the number of companies. It is in this way because while the companies were analyzed, a particular management structure in a South Korean company indicates that the company is run by three CEOs, who are in charge of making decisions in the company. Given this particularity, these CEOs were taking into account for the present study.

Finally, the number of companies and CEOs analyzed are (Table 1):

To achieve our general objective established, the following three research questions were defined:

RQ1: Do CEOs of companies have a presence in SNS? Do they have a presence on Twitter? (Presence)

RQ2: What activity do CEOs develop on Twitter? (Activity)

RQ3: What kind of interaction is developed on CEOs Twitter accounts? (Interaction).

In order to answer each research question, it has been designed different categories of analysis.

For answer RQ1 (the presence of CEOs in SNS), five of them have been selected: Facebook, Twitter, LinkedIn, Instagram and YouTube (Table 2). For doing this selection, the Digital in 2017 Global Overview report from We Are Social and Hootsuite has been considered (We Are Social & Hootsuite, 2017). In this report, the SNS selected for the analysis are classified as the most active social media platforms and the ones with the greatest number of active users.

Once the five SNS were selected, we went into them and look for the accounts of the CEOs that were identified from both rankings. The collected information was sort in a table and classified according to the name of the CEO and if each one of them has an account in any SNS.

For analyzing the Activity of CEOs on Twitter (RQ2), we study if the accounts were active or inactive. Based on different studies (Collier, 2014; González Pimentel, 2016; Patel, 2016; SocialBakers, 2013; Skaf, 2016), it was established several levels of Intensity of tweeting (Table 3). According to these reports, the adequate intensity of tweeting would be between 3 and 5 tweets per day (around 25 and 35 tweets per week).

Then, it was also looked for the type of tweet that CEOs write (Table 4). Thus, the tweets were classified in three groups: Tweet (TW), Retweet (RT) and Reply (RE). In the case of RE, we only considered the replies made by the CEO and not the replies that the tweet receives. It was done in that way because of our objective with this category of analysis was to understand the type of post from CEOs and determine which the activity developed in this SNS is.

For doing the analysis of the Interaction (RQ3), two dimensions were selected: (a) Interaction Level, and (b) Interaction Rate (Table 5). The Interaction Level made it possible to know the quantity of

Table 1
Sample of CEOs.

	Total		FG 500		AE 500	
	N°	%	N°	%	N°	%
Total companies from the ranking	1000	**100**	500	**100**	500	**100**
Total Analysis companies	985	**98.5**	490	**98.0**	495	**99.0**
Total CEOs	938	**93.8**	493	**98.6**	445	**89.0**

Table 2
Presence in SNS.

Category	SNS	Number of active users (in millions)
Presence	Facebook	1871
	YouTube	1000
	Instagram	500
	Twitter	317
	Linkedin	106

Table 3
Intensity of the Activity.

Category	Dimension	Options
Activity	Intensity	0 tweets (Inactive) Less than 1 tweet per week (Quasi Inactive) Between 1 and 15 tweets per week (very low) Between 15 and 25 tweets per week (low) Between 25 and 35 tweets per week (adequate) Between 35 and 45 tweets per week (high) More than 45 tweets per week (very high)

Table 4
Type of Activity.

Category	Dimension	Options
Activity	Type of post	Tweet (TW) Retweet (RT) Reply (RE)

Table 5
Interaction.

Category	Dimension	Options
Interaction	Interaction Level	N° Retweets/N° Tweets N° Favorites/N° Tweets
	Interaction Rate	(N° Retweets/N° Tweets/N° Followers) × 100 (N° Favorites/N° Tweets/N° Followers) × 100

interaction, obtaining the number of interactions per post of each CEO. It is defined dividing the total number of Retweets and Favorites that CEO's Tweets received, in relation to the number of tweets posted in the period of time analyzed. The Interaction Rate allows us to know the quality of interaction, linking the volume of interactions with the number of followers. It is obtained by assessing the relationship between the quantity of Retweets and Favorites obtained per each tweet, in relation to the number of followers of the CEOs, and multiplying it for 100.

It was analyzed the tweets posted by CEOs during 2016. The period selected was 26 weeks (six months), from July 1, 2016, until December 31, 2016.

The information for RQ1 was collected in December 2016. The information for answering RQ2 and RQ3 was collected in February 2017. In order to obtain the tweets, it had been used Twitonomy (http://www.twitonomy.com). This online tool allows download the last 3000 tweets from any account and gives reports about the category of the tweet (it means if it is a tweet, retweet or reply) and the retweet count and favorite count of each one of them. It also gives the URL to link automatically to that tweet. In addition, it allows researchers to download the tweets in an Excel document.

All the information obtained was classified and coded in Excel tables created specifically for this research, in order to obtain the following results.

4. Results

The total quantity of tweets analyzed were 14,153 (4421 from the Global CEOs and 9732 from the Latin American CEOs). The results obtained, based on the research questions established previously, are presented below.

4.1. The presence of CEOs in social networks sites

As we can see in Table 6, only one in three CEOs has a presence in at least one SNS. Compared with the 90% of Fortune 500 companies with an account in SNS (according to DiStaso & McCorkindale, 2013), the presence of CEOs in social media is really very low. Latin American CEOs perform better than their Global colleagues since almost half of the Latin American CEOs have at least one SNS account while only the 26% of Global CEOs have one.

LinkedIn is by far the most elected SNS by CEOs (almost 90% of them), followed by Twitter (around one in four CEOs) and Facebook (10% of them). CEOs have a very reduced presence on Instagram and YouTube (less than 2%). Comparing Global and Latin American CEOs, it can be observed that both groups have a similar presence on LinkedIn, but a higher percentage of Global CEOs have presence on Twitter and Facebook than the Latin American ones: around 40% of Global CEOs have a Twitter account against 17% of Latin American ones, and on Facebook, 7% of Latin American CEOs have an account while the Global CEOs duplicate this percentage.

Table 6
Presence of CEOs on SNS.

	Total		FG 500		AE 500	
Presence in SNS	N°	%	N°	%	N°	%
Yes	324	**34.5**	128	**26.0**	196	**44.0**
No	614	**65.5**	365	**74.0**	249	**56.0**
Presence per SNS						
Linkedin	281	**86.7**	106	**82.8**	175	**89.3**
Twitter	86	**26.5**	53	**41.4**	33	**16.8**
Facebook	32	**9.9**	18	**14.1**	14	**7.1**
Instagram	6	**1.8**	2	**1.6**	4	**2.0**
YouTube	3	**0.9**	1	**0.8**	2	**1.0**

In relation to the presence of CEOs on Twitter, results show that the presence is low (around 25% of them), but Global CEOs have a more significant presence than Latin American ones: while almost half of the Global CEOs have a Twitter account, only less than 20% of Latin American CEOs have presence in this microblogging platform.

4.2. The activity of CEOs on Twitter

Once the tweets were downloaded, the accounts were divided into two groups: (a) Inactive: the ones that do not have any activity in the period studied, and (b) Active: the ones that have at least a minimal activity in the period analyzed.

The results show that a quarter of CEOs Twitter account is inactive (Table 7). This could be considered a high percentage, taken into account that only the 25% of CEOs have a Twitter account. The great majority of Latin American CEOs have active accounts, but only two-thirds of the Global CEOs are active on the platform.

Even though three of four CEOs have an active account, it is observed that 30% of them are quasi-inactive (less than 1 tweet per week), and 92% of active accounts have a very low intensity of activity (less than 25 tweets per week). Comparing Global and Latin American CEOs, the tendency is similar but Global CEOs perform worse than LatAm ones: 33% against 23% of quasi-inactive accounts, and 95% against 86% of low activity (Table 8). These results clearly show that very few CEOs (less than 10% of them) are doing well their work on Twitter.

In Table 9 it is possible to observe that CEOs post, on average, around eight tweets weekly in the analyzed period. But the Latin American CEOs (AV = 12,5 posts) publish more than 2,5 times than the Global CEOs (AV = 4,7 posts).

Per types of tweets, approximately 60% of posts were original tweets (TW), almost 30% were retweets (RT) and they only do replies (RE) once a week. The Global CEOs used to post 20% more their own tweets than Latin American CEOs (72% against 51%), but the last ones tend to reply four times more than the first ones (18.4% against 4.3%). That is, the main activity of CEOs is oriented to disseminate their own tweets, instead of sharing tweets from other people or engage in replies/conversations of their own tweets.

Table 7
Activity of CEOs on Twitter.

Activity	Total		FG 500		AE 500	
	N°	%	N°	%	N°	%
Inactive Accounts	20	**23.2**	17	**32.1**	3	**9.1**
Active Accounts	66	**76.7**	36	**67.9**	30	**90.9**
Total	86	**100**	53	**100**	33	**100**

Table 8
Intensity of Activity of CEOs on Twitter.

Intensity of activity (actives only)	Total (66)		FG 500 (36)		AE 500 (30)	
	N°	%	N°	%	N°	%
Less than 1 tweet per week (Quasi Inactive)	19	**29.8**	12	**33.3**	7	**23,3**
Less than 15 tweets per week (very low)	36	**54.5**	21	**58.3**	15	**50.0**
Between 15 and 25 tweets per week (low)	5	**7.6**	1	**2.7**	4	**13.3**
Between 25 and 35 tweets per week (adequate)	3	**4.5**	1	**2.7**	2	**6.6**
Between 35 and 45 tweets per week (high)	0	**0.0**	0	**0.0**	0	**0.0**
More than 45 tweets per week (very high)	3	**4.5**	1	**2.7**	2	**6.6**
TOTAL	66	**100**	36	**100**	30	**100**

Table 9
CEO's Activity (per types of tweets).

	Total	TW		RT		RE	
		N°	%	N°	%	N°	%
General	**14,153**						
Weekly Average per CEO (66)	8,2	4,7	**57.3**	2,3	**28.0**	1,2	**14.7**
Fortune Global (FG) 500	**4421**						
Weekly Average per CEO (36)	4,7	3,4	**72.3**	1,1	**23.4**	0,2	**4.3**
America Economia (AE) 500	**9732**						
Weekly Average per CEO (30)	12,5	6,4	**51.2**	3,8	**30.4**	2,3	**18.4**

4.3. The interaction of CEOs on Twitter

The results from Table 10 show that CEO's tweets receive a mean of 80 Retweets and Favorites per each tweet. This is a high level of interactions, despite almost the 60% of these interactions are Favorites (less relevant than Retweets).

Despite the fact that Latin American CEOs post more tweets than the Global ones, we can observe a great difference in the interactions obtained. Global CEOs obtain almost seven times more interactions than LatAm CEOs (a mean of 190 interactions per tweet against 28 interactions). The tweets from the Global CEOs receive four times more Retweets than the LatAm ones (approximately 67 RT per tweet against 16 RT). If we look into the Favorites, the difference is even bigger: the Global CEO's tweets receive ten times more Favorites than Latin American CEO's tweets (near 125 F per tweet against 12 F).

The results from Table 11 show that the average (AV) of Followers for each CEO is 166,267. The Global CEOs tend to have three times more followers (AV = 233,251) than the Latin American CEOs (AV = 85,885). This data could decisively influence the quality of interactions obtained by Global and LatAm CEOs since a higher volume of followers usually tends to reduce the rate of interactions obtained per each tweet.

Analyzing the results obtained, we can affirm that the different interaction rates obtained by CEOs are extremely low. In general, the total interaction rate does not arrive at one interaction (Favorites plus Retweets) every 20 tweets (almost 0.05 interactions per tweet), and the favorites' interaction represents almost the 60% of the total interaction rate. If the comparison is done between the Global and LatAm CEOs, we find that the first ones generate higher rates of interaction than the second ones (2.5 times more, 0.082 against 0.033). However, in both cases, the numbers are strongly low. It is relevant to point out that the Global CEOs have a higher volume of favorites interactions (65% favorites and 35% retweets), following the general results, but, on the contrary, the LatAm CEOs have a higher volume of retweets interactions (43% favorites and 57% retweets).

Table 10
Interaction Level.

Average per CEO (Weekly)	Interaction level						
	N° Tweets	Total (RT + F)		Retweets (RT)		Favorites (F)	
	N°	N°	AV/tw	N°	AV/tw	N°	AV/tw
General (66)	8,2	652,3	**79,5**	264,3	**32,2**	388,0	**47,3**
Fortune Global (FG) 500 (36)	4,7	899,8	**191,4**	315,9	**67,2**	583,9	**124,2**
America Economia (AE) 500 (30)	12,5	355,3	**28,4**	202,4	**16,2**	152,9	**12,2**

5. Conclusions

After analyzing the results obtained about the presence of CEOs on SNS, and the activity and interaction on their Twitter accounts, our research indicates some clear conclusions:

In relation to the presence of CEOs on SNS, it can be affirmed these results represent a big contrast to the general thought about the proliferation of the use of social media, at least in terms of CEOs of companies. One-third of the CEOs have no presence in any SNS. LinkedIn (with almost 90% of CEOs having an account on it) keeps being the most elected one and it could be understood as if CEOs are looking for a channel to show their profile and keep in touch with "professionals" and not to be in a direct contact with the general public. But it is striking that two of current key social media platforms like Facebook or Twitter have a very low presence of CEOs, although Global CEOs have a better presence than LatAm CEOs in both of them. So, against the commonly shared idea in communication management that CEOs should have a "voice" on social networks, our results show that nowadays they have a very poor presence on them.

Concerning to the activity developed by CEOs on Twitter, it could be suggested that they are not doing a good use of this social network as a communication tool. If we consider that only 25% of CEOs have a Twitter account, and almost 25% of them are inactive, and 90% of the active accounts have a very low activity, we can affirm that CEOs are not doing well on Twitter. Only 6 of 938 CEOs (the 0,64% of them) are doing the right thing on Twitter. Despite this fact, it is necessary to highlight that the Latin American CEOs are much more active than the Global CEOs. Besides this, taking into account the number of retweets and replies done in comparison with the numbers of tweets posted, the majority of CEOs keep mainly using Twitter as a one-way channel to disseminate their opinions and thoughts, not taking full advantage of the dialogical features of this platform.

With regard to the interaction results, it is possible to determine that CEOs have a high level of interactions, receiving a great number of Favorites and Retweets (more than 600 interactions per tweet), but if it is analyzed the Interaction rate obtained (considering the level of interaction in relation to the number of followers), we can conclude that the CEOs have a very low interaction rate. The very low Interaction rates tend also to confirm the idea expressed previously that Twitter is mainly used to disseminate their own ideas or information, but it is not used to engage in dialogue or to interact with their followers. On the other hand, Global CEOs obtain better interaction results than LatAm CEOs, not only in quantity of interactions (Interaction Level) but also in the Interaction Rate. In this way, these results may contradict a general statement associated with social media communication management: the bigger the number of followers, the lower the interaction rate. Here we can observe that, on average, Global CEOs have 2.5 times more

Table 11
Interaction Rate.

Average per CEO (Weekly)	N° Followers	Interaction rate N° Tweets	Total (RT + F)		Retweets (RT)		Favourites (F)	
	(average)	(average)	N	AV/tw	N	AV/tw	N	AV/tw
General (66)	166,267	8,2	652,3	**0.0478**	264,3	**0.0194**	388,0	**0.0284**
Fortune Global (FG) 500 (36)	233,251	4,7	899,8	**0.0820**	315,9	**0.0288**	583,9	**0.0532**
America Economia (AE) 500 (30)	85,885	12,5	355,3	**0.0330**	202,4	**0.0188**	152,9	**0.0142**

interaction rate than LatAm CEOs, although the first ones have 3 times more followers than the second ones. Therefore, it seems that people following Global CEOs tend to be more active than the ones following LatAm CEOs, or the first ones may be are doing a better work, publishing more interesting topics or motivating better their followers than LatAm ones do.

To sum up, the trend is growing at a very slow pace: the number of CEOs with a presence in SNS is increasing, but it is still very low. The same flow can be identified for both samples of CEOs studied, with the particularity that Latin American CEOs are a bit more active than the Global CEOs, but the second ones have better interaction results. Despite the growing presence and activity of CEOs, it seems that Twitter is not broadly being used by them as a dialogical tool, but mainly as a one-way tool to disseminate their own points of view.

In this way, it seems clear that the CEO presence and communication is not adequately managed on social media, and this is a field of strategic communication that should be improved in the next years by companies and by the CEOs themselves, to take advantage of a relevant and valuable tool to communicate and engage in dialogue with their publics.

Finally, we would like to look at the contributions and limitations of this study. This work presents a specific methodology suitable for analyzing the presence, activity, and interaction of different subjects (people and organizations) on Twitter, and it was specifically applied to a large group of CEOs of companies, both Global and Latin American ones. From an academic point of view, this work can be used by other researchers to analyze other samples of CEOs, managers or practitioners, and to promote the comparison of results. From a practitioner's perspective, the results of this study can also contribute to identifying some strengths and weaknesses of the CEO's digital communication strategies and to help in improving their communication management. However, the results can not be extrapolated to another kind of managers or even their companies. Thus, a possible future line of work could apply the methodology developed in this research to other types of subjects.

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